

Chapter – 5

Advanced Encryption Standard

(AES)

Recall weakness of the DES

- All of the cryptographic algorithms we have looked at so far have some problem.
- The earlier ciphers can be broken with ease on *modern computation* systems.
- The DES algorithm was broken in 1998 using a system that cost about \$250,000.
- It was also far too slow in software as it was developed for mid-1970's hardware and does not produce efficient software code.
- Triple DES on the other hand, has three times as many rounds as DES and is correspondingly slower.
- **What was required?** *A brand new encryption algorithm*

Inner Workings of a Round

Step 1: *State array construction and Key expansion*

- A single 128 bit block both for en/decryption is called *in matrix*
- The in matrix block is grouped into *16 bytes* and copied into a *state array (4 x 4 matrix)* which is modified at each stage of an algorithm & then copied to an *output matrix*.
- Both the plaintext & the encryption key are depicted as *128 bit square matrix of bytes*.
- The encryption key is expanded into an array of *key schedule words (the w matrix)*.
- It must be noted that the ordering of bytes within the in matrix is by column. The same applies to the w-matrix.

Step 2: *Add round key (making XOR)*

- The algorithm begins with an *Add round key* stage followed by 9 rounds of four stages and a tenth round of three stages.
- The 128 bit block XORed with the encryption key

Step 3: *Round operations*

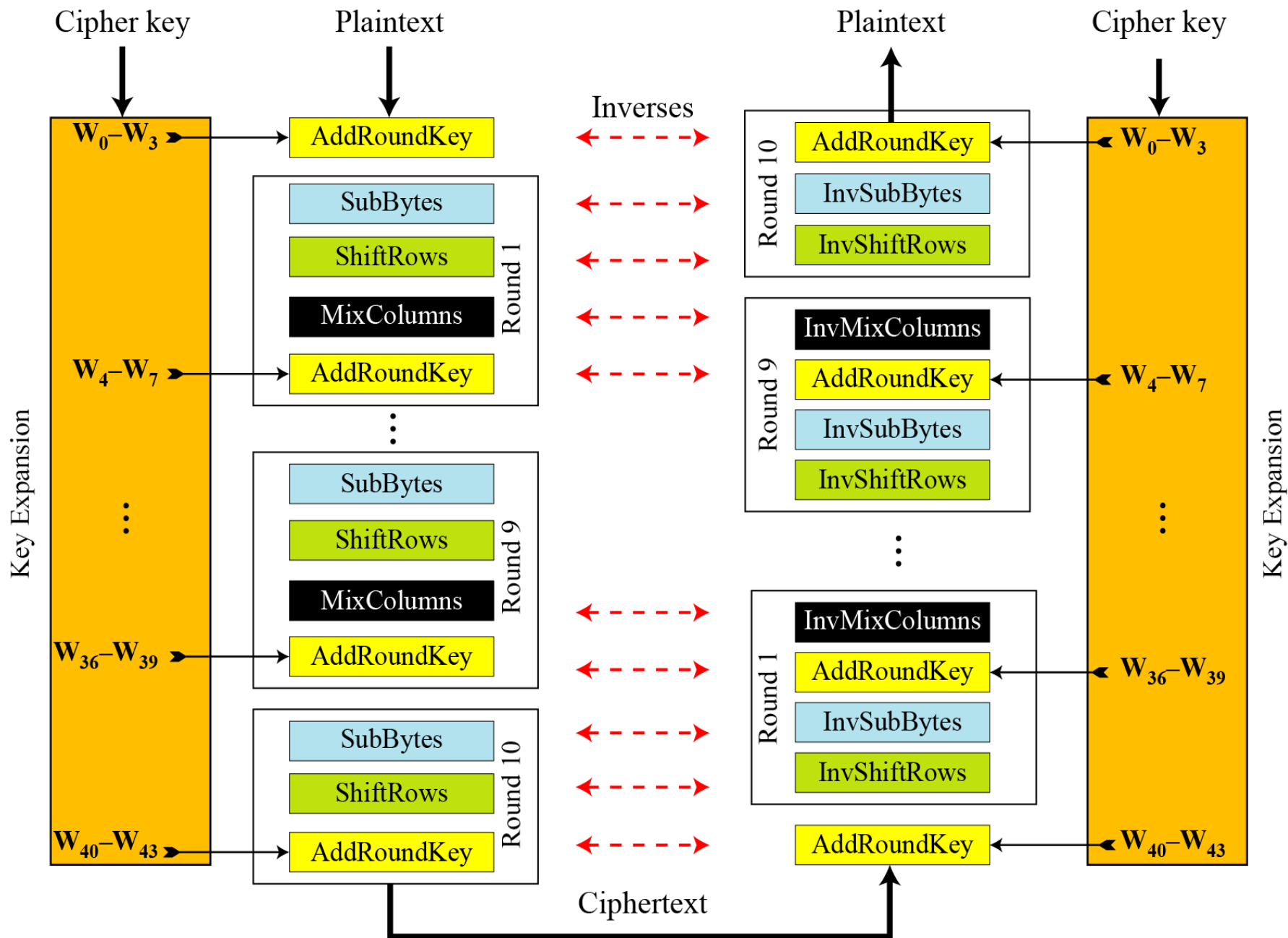
3.1. *Substitute bytes*

3.2. *Shift rows*

3.3. *Mix Columns (not used in the last round, 10th round)*

3.4. *Add Round Key*

Output --- Ciphertext



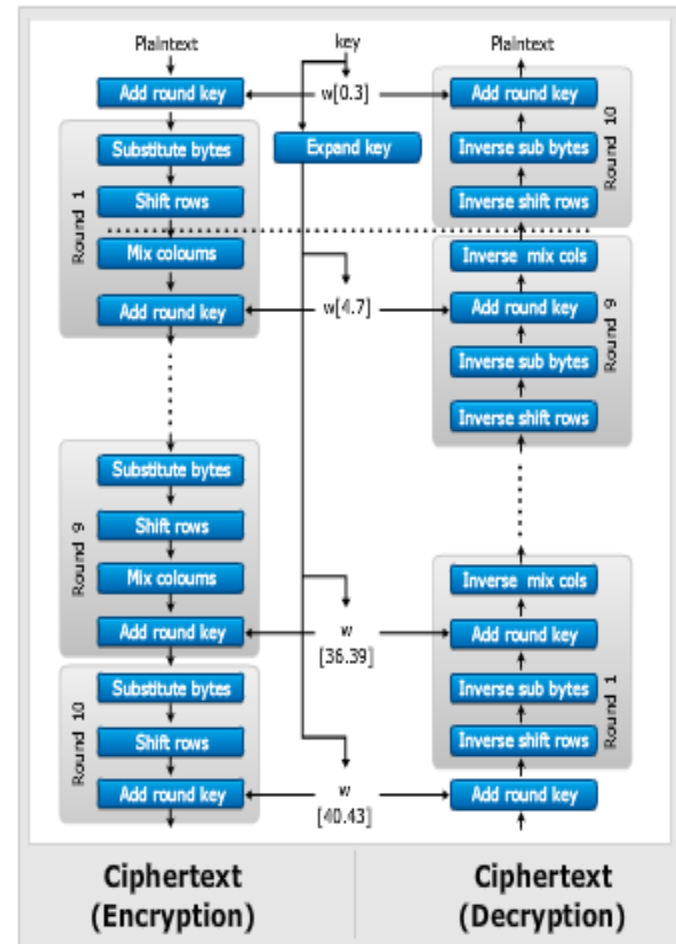
Advanced Encryption Standard (AES)

AES Algorithm

- Triple DES is more secure with its 168-bit key but it is slow in terms of software
- It is processed for 48 rounds and operates on 64-bit blocks.
- For security and efficiency reasons, a larger block size is preferred (at least 128 bits)
- National Institute of Standards and Technologies (NIST) issued a call for proposals for Advanced Encryption Standard (AES) in 1997

NIST specified the following requirements for AES:

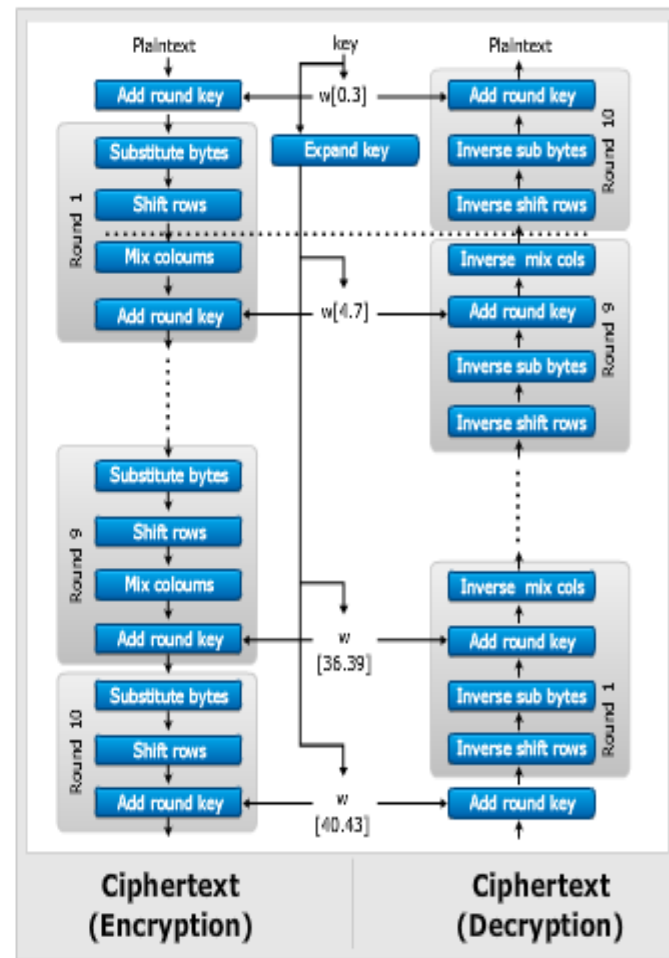
- It must be symmetric block cipher
- Block length is of 128 bits
- Key lengths are of 128, 192 and 256



Advanced Encryption Standard (AES)

NIST specified the following requirements for AES:

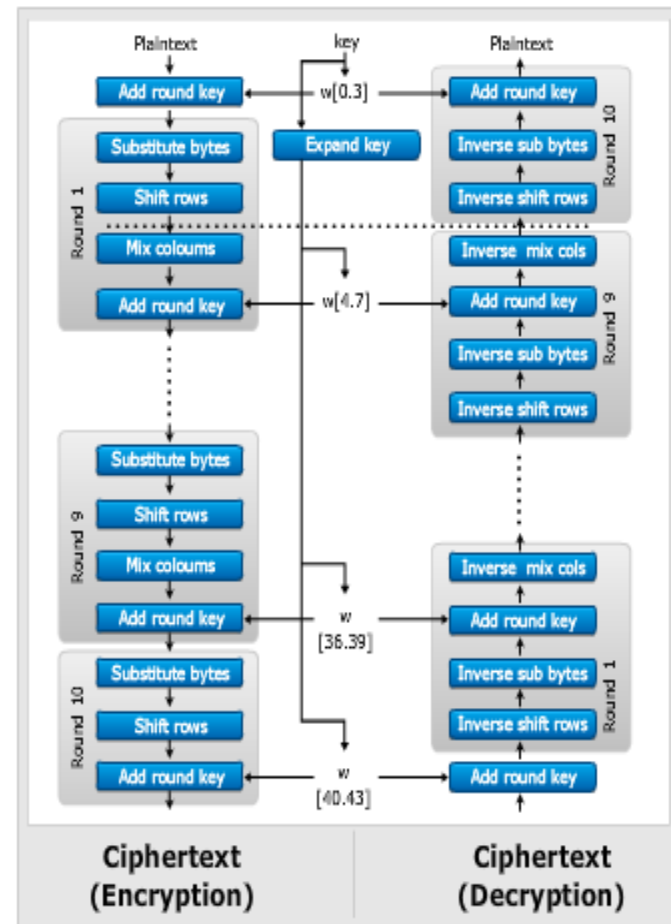
- Evaluation criteria include security, computational efficiency, memory requirements, hardware and software suitability and flexibility
- NIST selected Rijndael proposal as the proposed AES algorithm and published a final standard in November 2001
- Two cryptographers from Belgium, Dr. Joan Daeman and Dr. Vincent Rijmen developed the Rijndael proposal



Advanced Encryption Standard (AES)

AES Algorithm

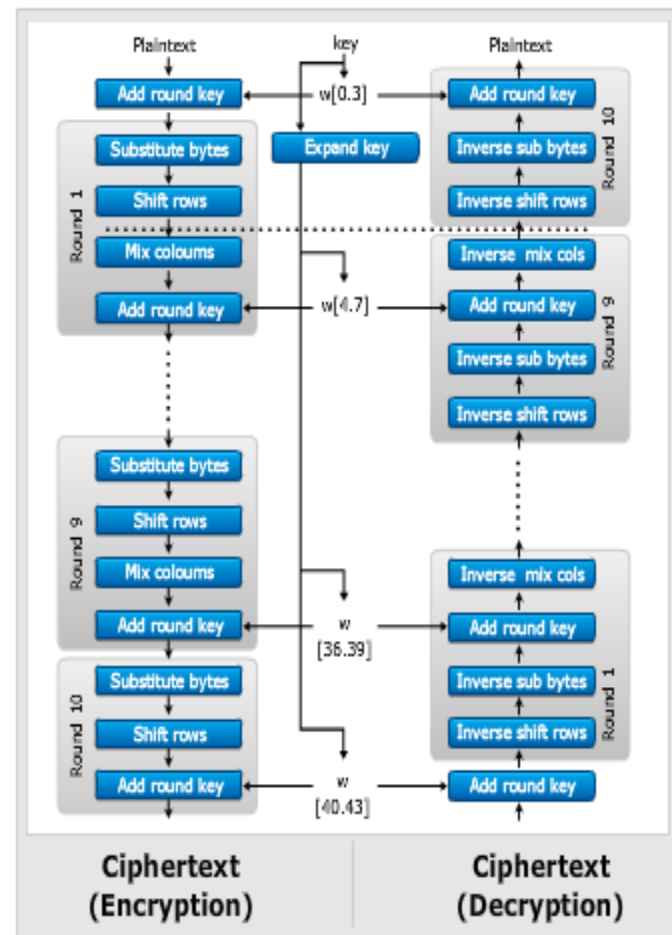
- AES uses a 128 bits block and key length of 128, 192 or 256.
- In general 128-bit key is used.
- The 128-bit block is copied into a state array.
- The 128-bit key is represented as a square matrix of bytes and the key is expanded to an array of 44 words.
- Four words are used as key for each round.
- AES operates in 10 rounds and four different stages are used in each round except 10th:
 - **Substitute bytes**: Uses S-box for substitution
 - **ShiftRows**: Simple permutation
 - **MixColumns**: Column-wise substitution
 - **AddRoundKey**: XORing the key with the block



Advanced Encryption Standard (AES)

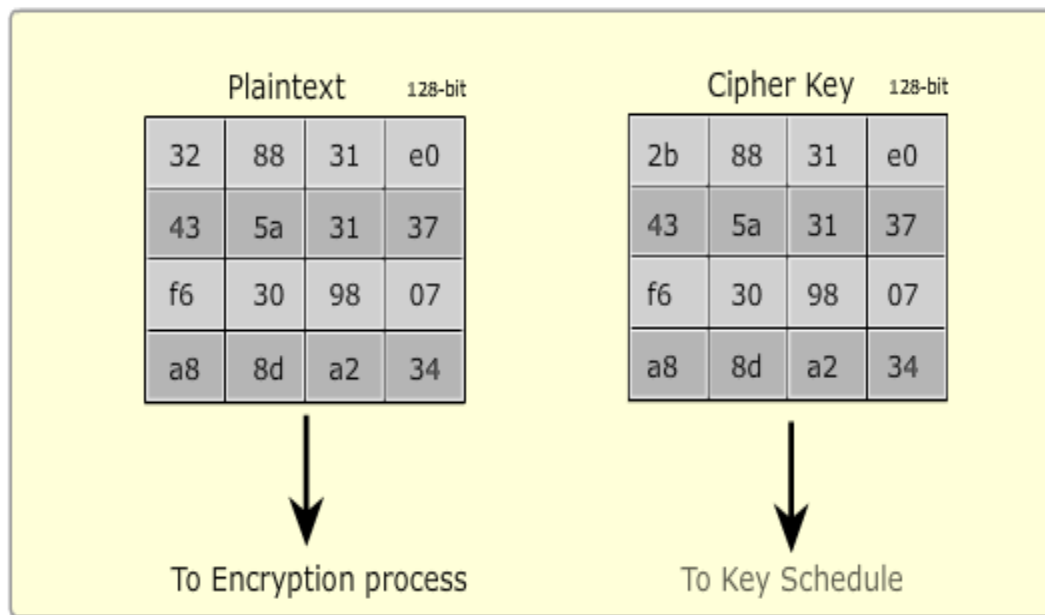
AES Algorithm

- The AES cipher begins with an AddRoundKey stage, followed by nine identical rounds, followed by the tenth round with three stages.
- AddRoundKey provides encryption. The other three stages provide confusion, diffusion and nonlinearity.
- Each round is reversible.

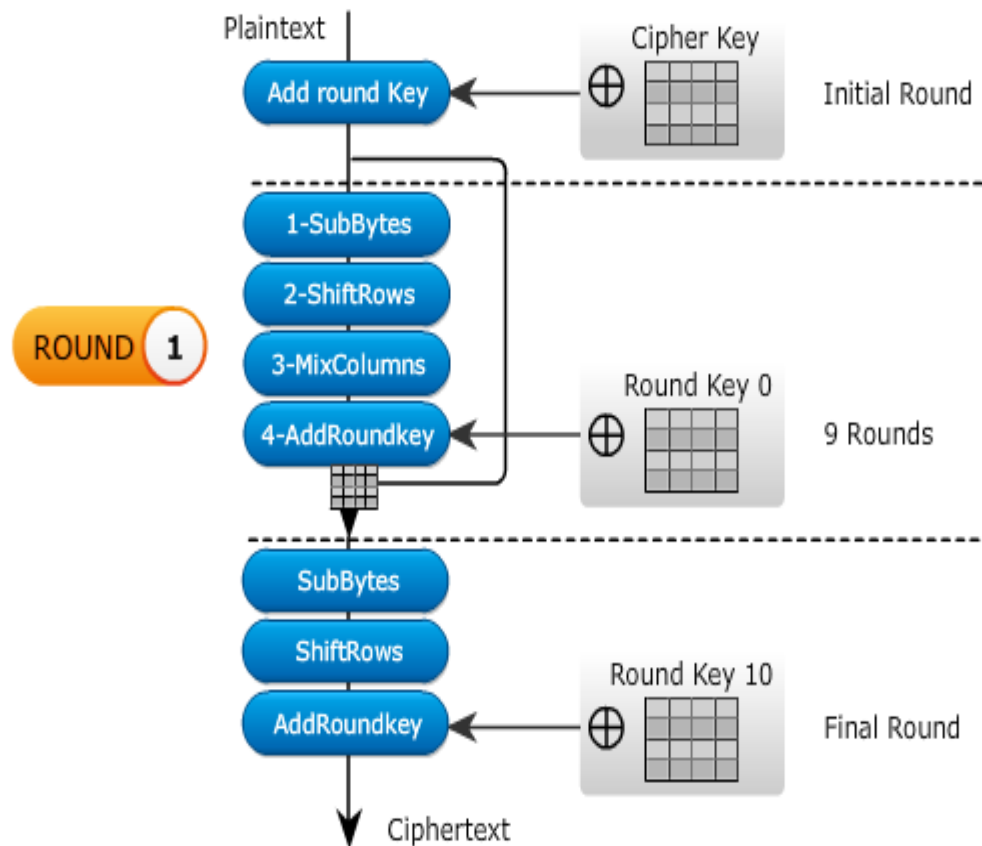


Advanced Encryption Standard (AES)

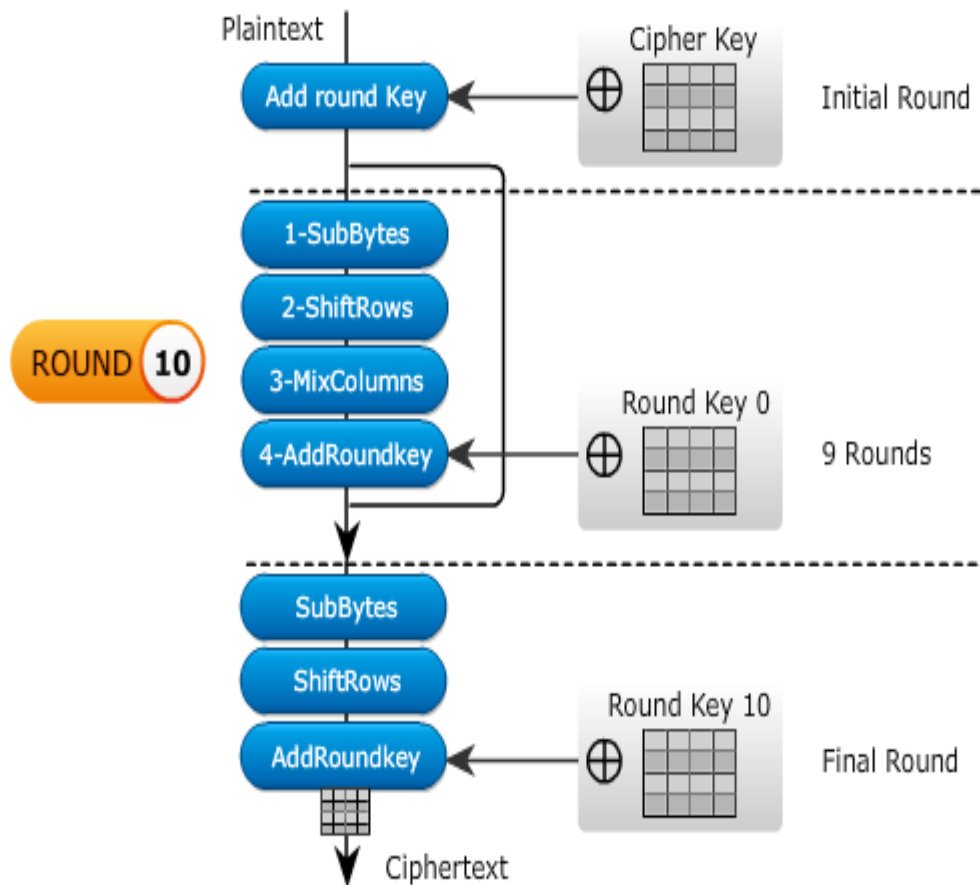
AES Alogarithm



Advanced Encryption Standard (AES)



Advanced Encryption Standard (AES)



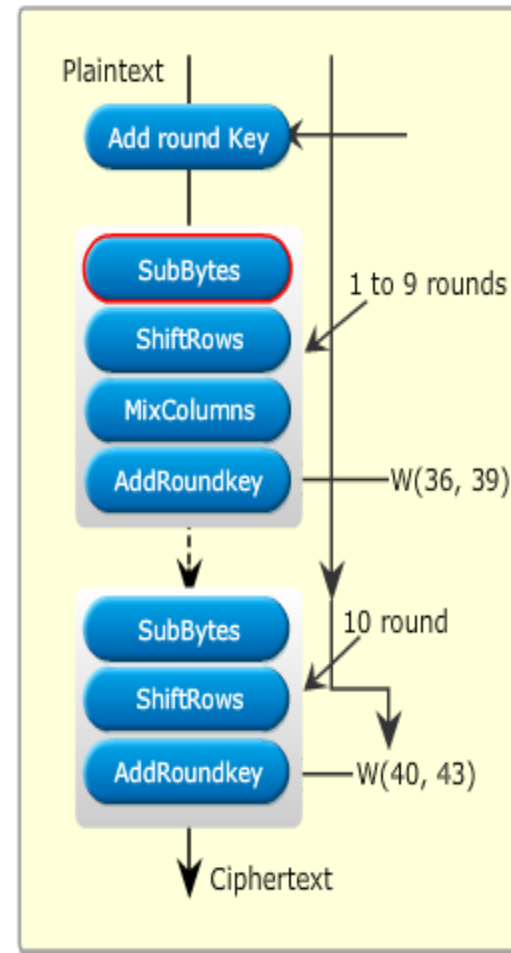
Advanced Encryption Standard (AES)

1-SubBytes

hex	y															
	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
0	63	7c	77	7b	f2	6b	6f	c5	30	1	67	2b	fe	d7	ab	76
1	ca	82	c9	7d	fa	59	47	f0	ad	d4	a2	af	9c	a4	72	co
2	87	fd	93	26	36	3f	f7	cc	34	a5	e5	f1	71	d8	31	15
3	4	c7	23	c3	18	96	5	9a	7	12	80	e2	eb	27	82	75
4	9	83	2c	1a	1b	6e	5a	a0	52	3b	d6	83	29	e3	2f	84
5	53	d1	0	ed	20	fc	81	5b	6a	cb	8e	39	4a	4c	58	c7
6	d0	ef	aa	fb	43	4d	33	85	45	f9	2	7f	50	3c	9f	a8
7	51	a3	40	8f	92	9d	38	f5	8c	86	da	21	10	ff	f3	d2
8	cd	0c	13	ec	5f	97	44	17	c4	a7	7e	3d	64	5d	19	73
9	60	81	4f	dc	22	2a	90	88	46	ee	b8	14	de	5e	0b	db
a	e0	32	3a	0a	49	6	24	5c	c2	d3	ac	62	91	95	e4	79
b	e7	c8	37	6d	8d	d5	4e	a9	6c	56	f4	ea	65	7a	ae	8
c	ba	78	25	2e	1c	a6	b4	c6	e8	dd	74	1f	4b	bd	8b	8a
d	70	3e	b5	66	48	3	f6	0e	61	35	57	b9	86	c1	1d	9e
e	e1	f8	98	11	69	d9	8e	94	9b	1e	87	e9	ce	55	28	df
f	8c	a1	89	0d	bf	e6	42	68	41	99	2d	0f	b0	54	bb	16

19	A0	9a	e9
3d	F4	C6	f8
E3	E2	8d	48
Be	2b	2a	08

d4			

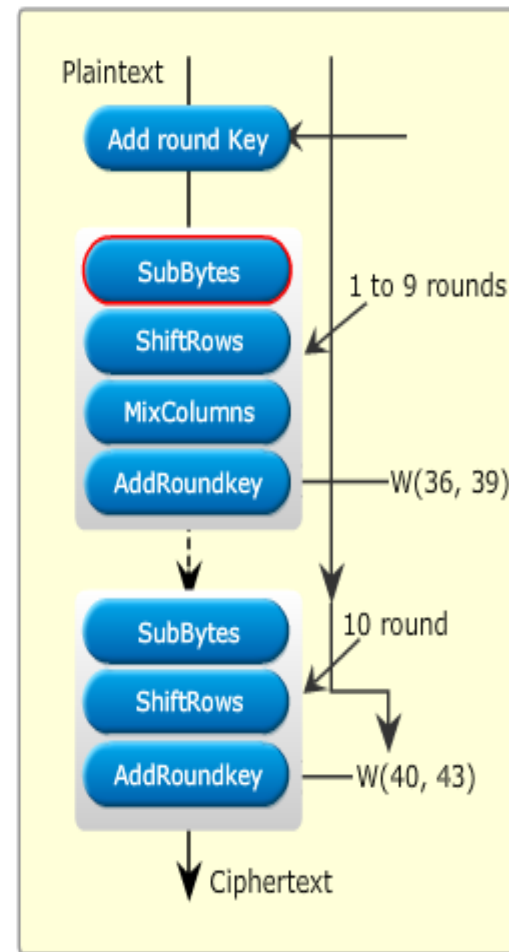


Advanced Encryption Standard (AES)

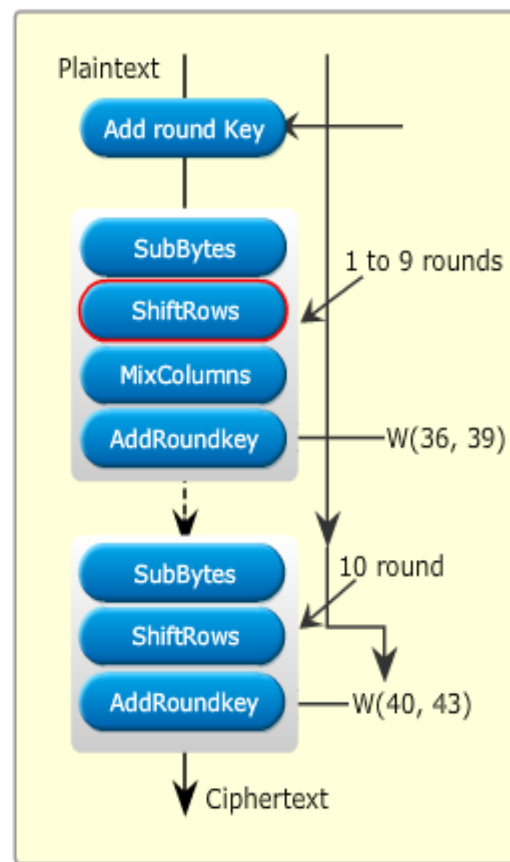
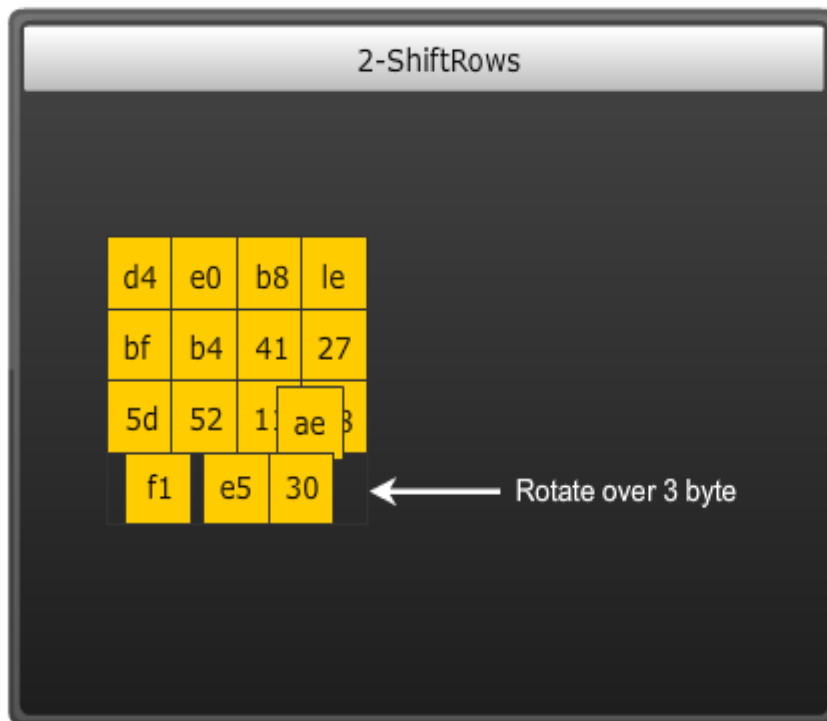
1-SubBytes																
hex	y															
x	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
0	63	7c	77	7b	f2	6b	6f	c5	30	1	67	2b	fe	d7	ab	76
1	ca	82	c9	7d	fa	59	47	f0	ad	d4	a2	af	9c	a4	72	co
2	87	fd	93	26	36	3f	f7	cc	34	a5	e5	f1	71	d8	31	15
3	4	c7	23	c3	18	96	5	9a	7	12	80	e2	eb	27	82	75
4	9	83	2c	1a	1b	6e	5a	a0	52	3b	d6	83	29	e3	2f	84
5	53	d1	0	ed	20	fc	81	5b	6a	cb	be	39	4a	4c	58	c7
6	d0	ef	aa	fb	43	4d	33	85	45	f9	2	7f	50	3c	9f	a8
7	51	a3	40	8f	92	9d	38	f5	8c	86	da	21	10	ff	f3	d2
8	cd	0c	13	ec	5f	97	44	17	c4	a7	7e	3d	64	5d	19	73
9	60	81	4f	dc	22	2a	90	88	46	ee	b8	14	de	5e	0b	db
a	e0	32	3a	0a	49	6	24	5c	c2	d3	ac	62	91	95	e4	79
b	e7	c8	37	6d	8d	d5	4e	a9	6c	56	f4	ea	65	7a	ae	8
c	ba	78	25	2e	1c	a6	b4	c6	e8	dd	74	1f	4b	bd	8b	8a
d	70	3e	b5	66	48	3	f6	0e	61	35	57	b9	86	c1	1d	9e
e	e1	fe	98	11	69	d9	8e	94	9b	1e	87	e9	ce	55	28	df
f	8c	a1	89	0d	bf	e6	42	68	41	99	2d	0f	b0	54	bb	16

19	A0	9a	e9
3d	F4	C6	f8
E3	E2	8d	48
Be	2b	2a	08

d4	e0	b8	1e
27	bf	b4	41
11	98	5d	52
ae	f1	e5	30

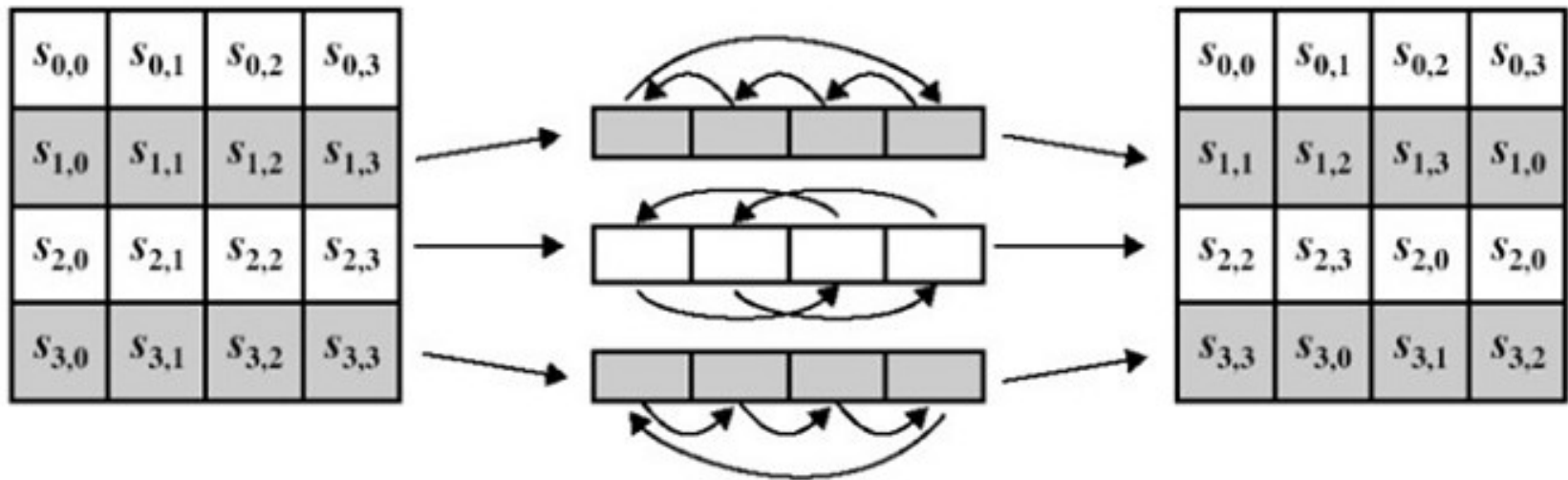


Advanced Encryption Standard (AES)



Shift Row Transformation

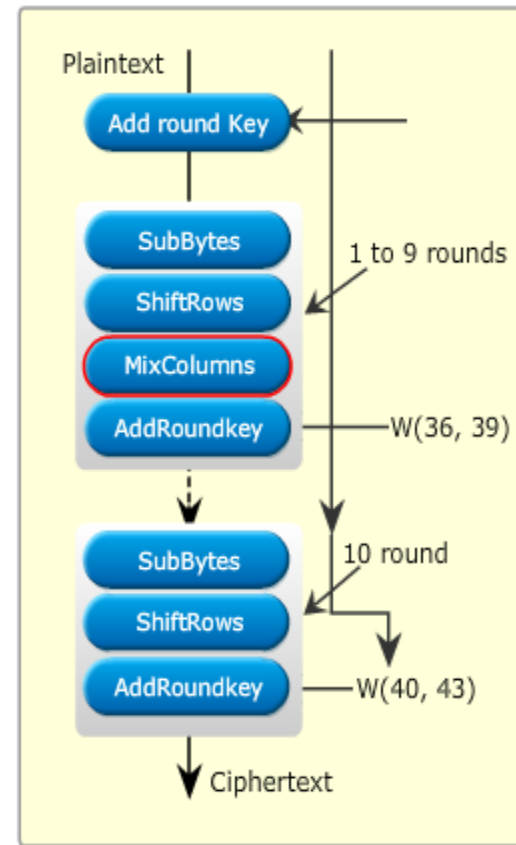
- It is a simple permutation which works as follow:
 - The 2nd row is shifted 1 bytes to the left in a circular manner.
 - The 3rd row is shifted 2 bytes to the left in a circular manner.
 - The 4th row is shifted 3 bytes to the left in a circular manner.



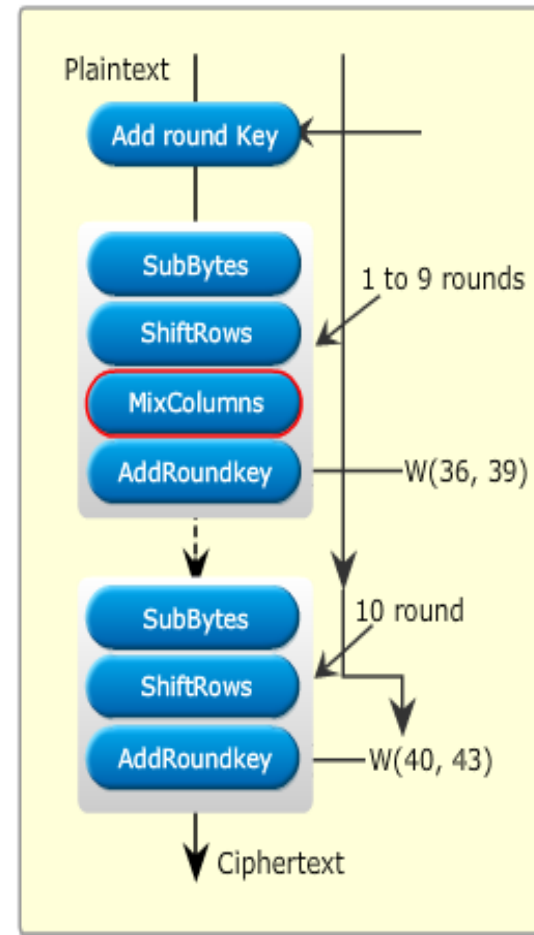
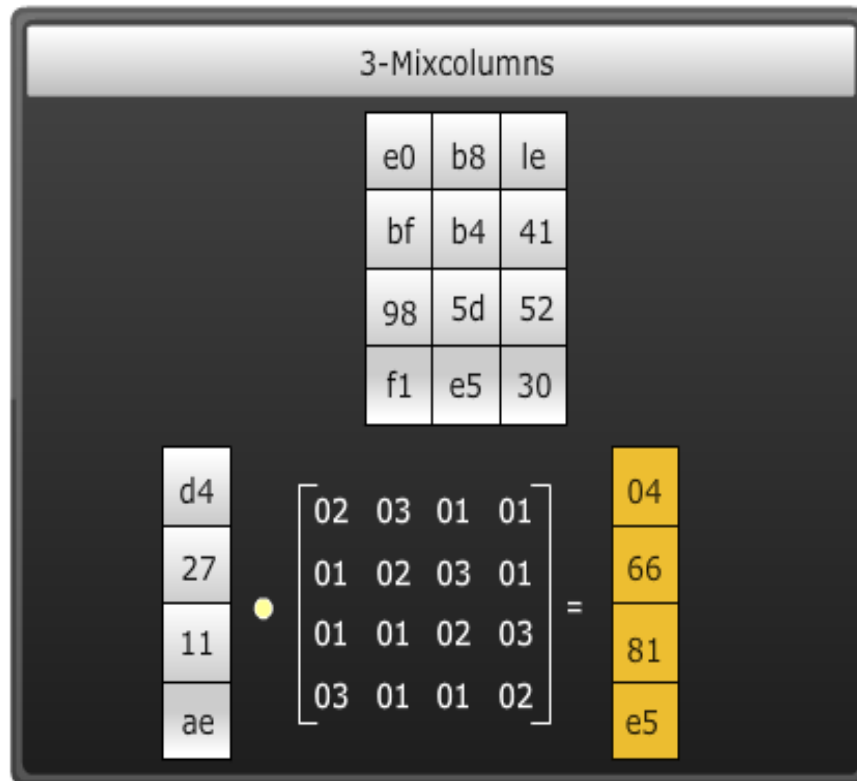
- The *Inverse Shift Rows transformation* (known as *InvShiftRows*) performs these circular shifts in the opposite direction for each of the last 3 rows (the first row was unaltered to begin with).

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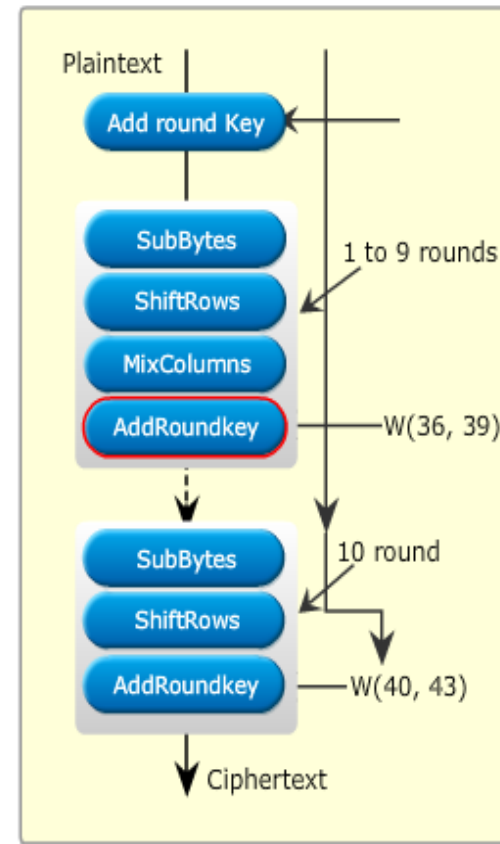
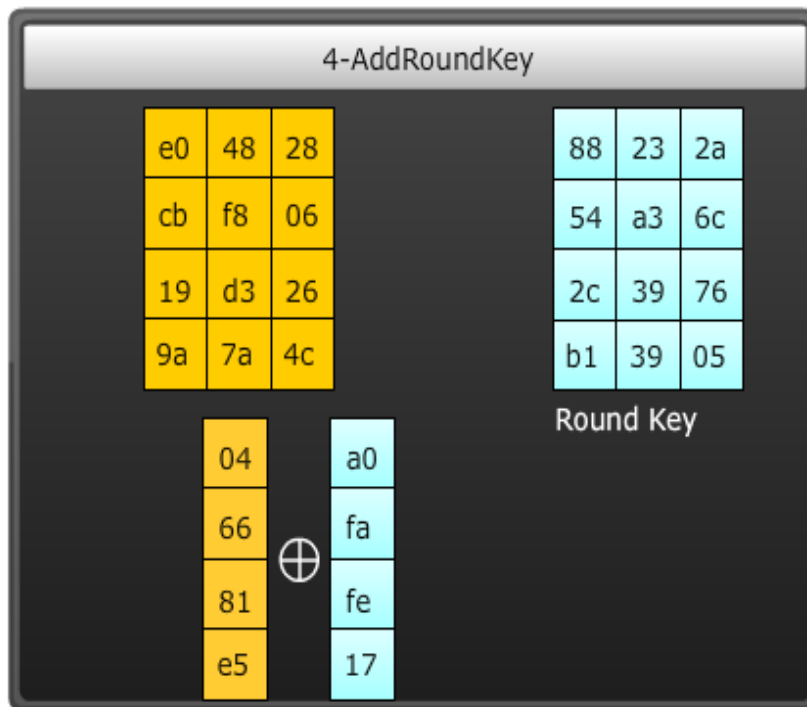
3-Mixcolumns			
d4	e0	b8	le
27	bf	b4	41
11	98	5d	52
ae	f1	e5	30



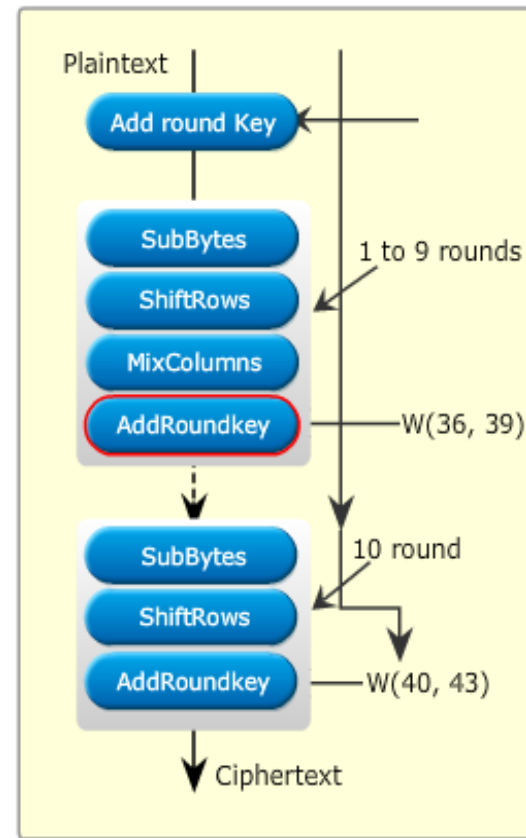
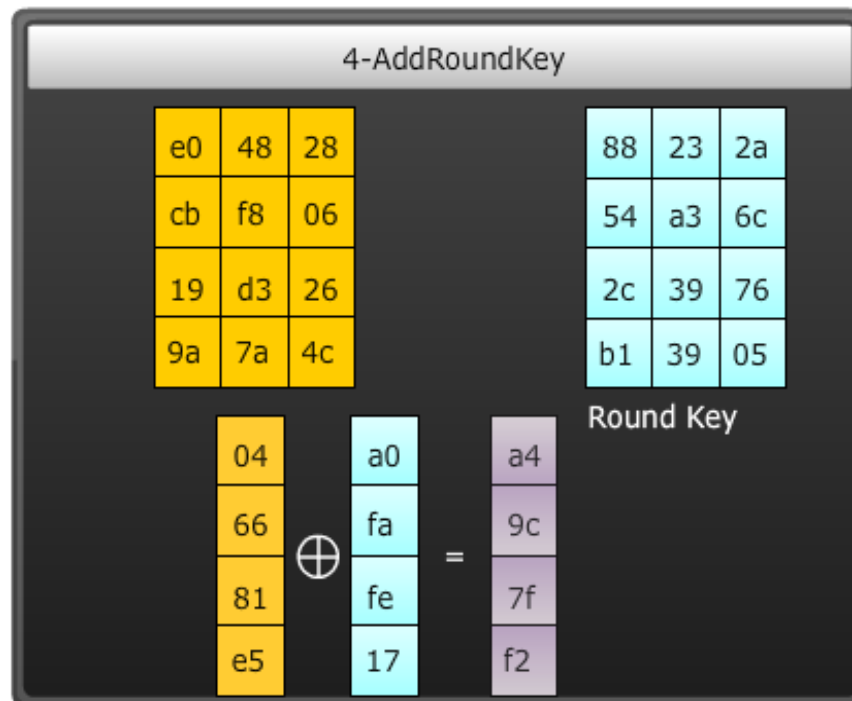
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After AddRoundKey

39	02	dc	19
25	dc	11	6a
84	09	85	0b
1d	fb	97	32

Ciphertext

